YASSER EL-SAYED M. EL-ALFY

Researcher | KFUPM

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Yasser El-Alfy is a researcher at KFUPM working on learning-based perception and 3D SLAM. His work focuses on integrating deep learning with traditional robotics to build robust navigation systems for autonomous vehicles and drones.

CITATIONS: 21 | H-INDEX: 2 | i10-INDEX: 1

EDUCATION

Doctor of Philosophy in Computer Engineering
King Fahd University of Petroleum and Minerals, Saudi Arabia | 2023-present

Master of Science in Computer Engineering
King Fahd University of Petroleum and Minerals, Saudi Arabia | 2021-2022

CGPA: 3.688

Bachelor of Science in Computer & Communication Engineering (BSCCE)
Ahlia University, College of Engineering, Bahrain | 2018-2020

CGPA: 3.99

High School
Ahlia Schools, King Fahd University of Petroleum and Minerals, Saudi Arabia | 2012-2014

Cumulative Grade: 99.92%

WORKSHOPS

Advanced Quantum Computing Workshop
Saudi Data and Artificial Intelligence Authority (SDAIA) | 11-15 Jan 2026

PROFESSIONAL EXPERIENCE

Teaching and Lab Assistant
King Fahd University of Petroleum and Minerals (KFUPM) | 2023-present

Research Intern
Center for Communication and Information Technology Research (CCITR), Research Institute, KFUPM | 2020

Teaching and Lab Assistant Ahlia University | 2018-2020

RESEARCH STRATEGY

My research focuses on bridging the gap between raw sensor data and high-level spatial understanding. I develop monocular depth estimation models and integrate them into SLAM pipelines to enable 3D mapping without expensive LiDAR hardware.

Key Focus Areas:

- Monocular Depth Estimation
- Visual SLAM & Navigation
- Road Anomaly Characterization
- Edge AI for Robotics
- 3D Mapping
- Scene Understanding

TECHNICAL SKILLS

Languages:	Java, JavaScript, Visual Basic, Python, TensorFlow, Pytorch, HTML, Assembly, Arduino, VHDL, Verilog
Software:	Matlab & Simulink, GNU Octave, Sumo Simulator, Cooja/Contiki Simulator, Xilinx, Modelsim, Cisco Packet Tracer, Arduino IDE, mikroc pro, ISIS 7 Professional, Eclipse, Visual Studio, Visual Studio Code, Google Colab, Jupyter Notebook, Microsoft office programs (Word, Excel, PowerPoint, Access), Latex professional editing, Overleaf, Adobe Photoshop, GIMP, Adobe Indesign, DaVinci Resolve, PowerDirector, Kdenlive, Windows Movie Maker, Microsoft Teams, Zoom, Google Meet, Cisco Webex, Blackboard, Moodle
OS:	Background in different Operating Systems: Windows, Mac, Linux
Other Skills:	Teamwork and communications skills, Research skills, Video and graphical editing

TEACHING ACADEMIC PROFILE

ITCS 122: Introduction to Programming Techniques	2018-2020
ITCS 121: Computer Programming	2018-2019
ITCS 101: Introduction to Computers & IT	2018-2020
ITCS 222: Visual Programming	2018-2019
ITMS 325: Web Applications Design	2018-2019
ITCS 224: Data Structures	2019-2020
ITCS 201: Object Oriented Programming I	2019-2020
COE 301: Computer Organization	
COE 292: Introduction to Artificial Intelligence	
COE 412	

TALKS & EVENTS

Multi-Modal Curriculum Federated Learning for Breast Cancer detection Seminar | 26th November 2025, 4:00PM

Bldg. 22, Room 119, King Fahd University of Petroleum and Minerals (KFUPM)

CERTIFICATIONS

IELTS Certificate: 6.5 2020

ICDL Certificates in Microsoft Office (Word, Spreadsheets, PowerPoint) 2018

Certificates of appreciation for being lab assistant, Ahlia Univ. 2018

Qiyas - General Aptitude Test: 100% 2014

Qiyas - Achievement Aptitude Test: 97% 2014

SELECTED PUBLICATIONS

Enhancing pothole detection and characterization: integrated segmentation and depth estimation in road anomaly systems
Systems and Soft Computing (2025) (2025)

Monocular based 3D depth estimation and SLAM integration
Drone Systems and Applications (2025) (2025)

Intelligent Road Anomaly Detection with Real-time Notification System for Enhanced Road Safety
Sustainability (2025) (2025)

Real-Time On-Device Vision-Based Road Anomaly Detection with Frame Span Interval (FSI) Event Counting
IEEE AICCSA (2025) (2025)

Enhancing monocular depth estimation with an advanced encoder-decoder architecture
Neural Computing and Applications (2025) (2025)

Dynamic Real-time Learning of Electromagnetic Interference Parameters for Drone Crashing Attack
IEEE PIMRC (2025) (2025)

A Comparative Study of Transfer Learning and Custom Models with SSD for Face Mask Detection
IEEE (2025) (2025)

Pothole Detection Under Various Weather Conditions Using YOLOv9
Remote Sensing (2025) (2025)

A federated learning approach to banking loan decisions
ISNCC (2023) (2023)

Enhancing Monocular Depth Estimation Using an Enhanced Encoder-Decoder Model
Sensors (2023) (2023)

Distributed Intrusion Detection Systems Based on Deep Learning Techniques and Boosting Ensemble
ACM (2023) (2023)

Arduino based obstacle awareness vest for blind people
ICT-ASET (2022) (2022)

MONOCULAR DEPTH ESTIMATION USING DEEP LEARNING FOR 3D SLAM MAPPING IN AUTONOMOUS NAVIGATION

KFUPM Thesis (2022) (2022)

Enhancing Monocular Depth Estimation Using an Enhanced Encoder--Decoder Model
Authorea Preprints, 2023 (2023)

Enhancing monocular depth estimation with an advanced encoder-decoder architecture
Neural Computing and Applications 37 (29), 24265-24280, 2025 (2025)

PATENTS

SYSTEM AND METHOD FOR DETECTING AND COUNTING ROAD ANOMALIES US12445815

DATASETS

Road Anomaly Dataset with Depth Ground Truth
The first pothole dataset with depth ground truth values, manually annotated for high accuracy.

Pothole Detection Under Various Weather Conditions Dataset
A dataset with pothole images and generated/simulated various weather conditions.